

PRIZCO ZULVANO SHAPUTRA

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PROFILE

I am Prizco Zulvano Shaputra student majoring in Automation Engineering. In the field of electricity, I have learned more than 6 years from high school to college in assembling an electrical system. I have a leadership spirit and are able to work in teams.

EDUCATION

Politeknik Perkapalan Negeri Surabaya – Surabaya, Indonesia **Jul 2022 – Sep 2026**
Majoring in Automation Engineering, ***GPA 3.34 / 4.00***

SMKN 3 Boyolangu – Tulungagung, Indonesia **Aug 2019 – Aug 2022**
Electrical Engginering

WORK EXPERIENCES

PT Indo Trans Teknologi – Surabaya, Indonesia **Apr 2026 – Okt 2026**
Technical Support Engineer

- Monitor fuel consumption levels on ships using sensors such as Flow Meters.
- Monitor engine health, maximum RPM limits, and ship component performance for optimal performance and damage prevention.
- Accurately track ship positions, coordinates, and movements using Hybrid GPS Tracker technology.

CV ROV Indonesia – Surabaya, Indonesia **Jul 2025 – Des 2025**
Mechatronic Enggineer

- Contribute to the development of remotely operated vehicle brushes for cleaning ship hulls
- Applied Kaizen principles (5S & PDCA/SDCA) to improve workplace organization, standardization, and project effectiveness.
- Contribute to the development of a remotely operated vehicle for checking the thickness of ship hulls.
- Contribute to the development of a remotely surface vehicle for checking lake depths and monitoring.

SMP Al-Azhar – Surabaya, Indonesia **Nov 2024 – Jan 2025**
Robotic Coachs

- Instructed students in robotics, covering microcontroller programming.
- Trained students in programming logic, basic electronics, and robot mechanics for school-level robotics competitions.
- Mentored and supervised junior high school robotics teams participating in regional and national competitions.

CV.BANGKIT CCTV – Tulungagung, Indonesia **Jan 2020 - Jun 2022**

- Employed as a technical staff member specializing in the installation and maintenance of CCTV surveillance systems for individual and corporate clients.
- Conducted customer needs analysis and delivered tailored technology-based solutions.

PROJECTS

Implementation of ROS on ROV for Sonar-Based Distance Maintenance Using Sliding Mode Control (SMC) **Ags 2026**

- Developed a ROS-based control system for autonomous underwater distance keeping using real-time sonar measurements.
- Integrated sonar sensing and ROS communication nodes for real-time acquisition and processing of underwater distance data.
- Implemented a Sliding Mode Control (SMC) algorithm to regulate ROV position and maintain a predefined distance from underwater targets.
- Designed the closed-loop control architecture integrating sonar feedback, SMC, and ROV thruster actuation.
- Evaluated distance-tracking and control performance under varying underwater conditions.

Ship Fuel Consumption Monitoring System **Jul 2026**

- Monitored ship fuel consumption using Flow Meter sensors installed in the vessel's fuel monitoring system.
- Collected and analyzed real-time fuel consumption data to monitor fuel usage during ship operations.
- Assisted in identifying abnormal fuel consumption conditions based on sensor readings and operational data.
- Supported monitoring activities to improve fuel usage visibility and operational efficiency.

Ship Engine Health & RPM Monitoring System **Jun 2026**

- Monitored engine health parameters and maximum RPM limits to support safe and reliable ship operation.
- Observed engine and ship component performance through real-time monitoring data.
- Assisted in identifying abnormal operating conditions that could potentially lead to component damage.
- Supported technical monitoring activities to maintain optimal engine and equipment performance.

Hybrid GPS-Based Vessel Tracking System **Mei 2026**

- Monitored vessel position, coordinates, and movement using Hybrid GPS Tracker technology.
- Observed real-time vessel locations to support operational monitoring and tracking.
- Assisted in verifying vessel movement and location data obtained from the tracking system.
- Supported vessel monitoring activities by ensuring location and movement information was properly tracked.

ROV (Remotely Operated Vehicle) with 3-Camera Monitoring System **Jan 2026**

- Designed and developed a semi-autonomous and manual-controlled ROV system for underwater operations.
- Integrated three onboard cameras with three dedicated Ground Control Station (GCS) monitoring displays.
- Developed real-time video transmission system for multi-angle underwater observation.
- Designed control system architecture for stable maneuvering and operational reliability.

Computer Integrated System-SCADA **Apr 2025**

- Developed a module for the Computer Integrated System-SCADA to integrate and control industrial processes.

Spiral Wind Turbine Power Plant Monitoring System (IoT-Based) **Ags 2024**

- Developed an IoT-based monitoring system for a spiral wind turbine power generation system..
- Built a remote monitoring dashboard for system status visualization and data logging.
- Developed early fault detection to enhance turbine reliability and operational safety.

Ocean Wave Power Plant Monitoring System (IoT-Based) **Ags 2024**

- Developed an IoT-based monitoring system for an ocean wave power generation system.
- Implemented data logging and remote monitoring to analyze power output performance.
- Built a visualization dashboard for system status tracking and performance evaluation.

Neural Network-Based Fresh Water Pump Motor Monitoring System **Ags 2024**

- Developed an IoT-based monitoring system for ship fresh water pump motors using Neural Network methods.
- Designed real-time data acquisition for motor parameters (temperature, vibration, and current).
- Integrated monitoring dashboard for condition visualization and early warning system.
- Conducted power pack condition monitoring to ensure system reliability and operational efficiency.

Website-Based Monitoring And Preventive Maintance System For Powerpack Using Decision Tree Method **Dec 2023**

- Design and implement monitoring and preventive systems to predict before powerpack failure.
- integrate tools from hardware to software and incorporate methods to make them more effective..

ORGANIZATION EXPERIENCES

Marine Robotic Comunity - **Surabaya, East Java** **Jan 2023 – Apr 2026**

HYYDROSHIPS – Underwater Robot Team

Shipbuilding Institute of Polytechnic Surabaya Robotics Community

- Actively participated as a team leader in HYDROSHIPS robotics team developing underwater robots.
- Designed and built mechanical and electronic systems for underwater navigation and exploration robots.
- Programmed robot control systems to manage movement and sensor operations.
- Conducted testing and calibration of underwater control and sensor systems to enhance accuracy and performance.

Internship at CV ROV Indonesia - **Surabaya, East Java** **Sep 2025 - Dec 2025**

- ROV assembly, maintenance, and trials.
- Teaching students about technological innovation and robots.

Sekolah Robot Indonesia- **Surabaya, East Java** **Jan 2025 – Aug 2025**

Robotics Instructor & Coach

- Taught and mentored elementary school students in robotics, electronics, mechanics, sensors, microcontrollers, and programming through hands-on learning.
- Guided students through robot design, assembly, programming, testing, and troubleshooting, while developing their problem-solving and critical-thinking skills.

Automation Week VII - **Surabaya, East Java** **Jul 2024 - Dec 2024**

Event Division

National Student Competition in the fields of Robotics, Scientific Writing and PLC

- Collaborated in brainstorming and designing event concepts, scheduling, and outlining the overall event agenda.
- Provided on-site support during the event, overseeing smooth transitions between sessions, managing attendee registration, handling last-minute adjustments, and ensuring all event needs were met.

Kontes Kapal Indonesia 2024 - **Surabaya, East Java** **Oct 2024**

Event Competition Division

National Maritime-Shipping Technology Competition for Students by the Ministry of Education and Culture of the Republic of Indonesia

- Coordinated competition logistics including scheduling, participant documentation, and real-time contest supervision.
- Collaborated with fellow committee members to ensure smooth event execution and compliance with contest regulations.
- Assisted in technical evaluations and supported the judging and scoring process.

Graduation Caremony - **Surabaya, East Java** **Des 2024**

Shipbuilding Institute of Polytechnic Surabaya Robotics Community

- Appointed as the coordinator for the 2024 Automation Engineering Graduation Ceremony, overseeing all technical and non-technical aspects of the event.

Himpunan Mahasiswa Teknik Otomasi (HIMATO PPNS) - **Surabaya, East Java** **Mar 2024 - Dec 2024**

Department of Dagri Staff

- Responsible for coordinating internal organizational activities.
- Organized and supervised internal programs, including regular meetings, training sessions, and performance evaluations.
- Fostered teamwork and collaboration among members to improve internal communication and operational effectiveness.

ACHIEVEMENT

1st Place Winner MATE ROV ASEAN Singapore **Apr 2026**
Singapore American School in collaboration with MATE ROV

Top 10 Finalist Photo Challenge (International) **Dec 2025**
Mate ROV Competition

2nd Place Winner SAUVC Singapore **Mar 2025**
National University of Singapore (NUS) in collaboration with the Singapore Polytechnic

2nd Place Winner KRBAI in Kontes Robot Indonesia (KRI) National **Jun 2024**
Balai Pengembangan Talenta Indonesia (BPTI) Pusat Prestasi Nasional Puspresnas

Best Design KRBAI in Kontes Robot Indonesia (KRI) National **Jun 2024**
Balai Pengembangan Talenta Indonesia (BPTI) Pusat Prestasi Nasional Puspresnas

1st Winner KRBAI in Kontes Robot Indonesia (KRI) Regional **Jun 2024**
Balai Pengembangan Talenta Indonesia (BPTI) Pusat Prestasi Nasional Puspresnas

SKILLS

Language	**:** Indonesian, English (Passive)
Soft Skills	**:** Leadership, Effective Communication, Team Collaboration, Problem Solving, Adaptability, Critical Thinking.
Technical Skills	**:** Microsoft Office (Word, Power Point, Excel), Visual Studio Code, Autodesk Fushion, EAGLE, CX-Programmer (PLC Omron), CX-Designer, Arduino IDE, Proteus, Canva.